

FreDEA: Frequency-Aware Dual External Attention for Time Series Forecasting

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Abstract. The core challenge in long-term multivariate time series forecasting is the joint modeling of temporal dependencies and cross-variable interactions from historical observations. In real-world data, trends, periodic patterns, and noise are often strongly entangled, making purely time-domain modeling difficult to decouple these components effectively. Existing methods either rely on fixed decomposition or use intra-sample self-attention, which makes it difficult to reuse recurring temporal patterns across windows. To address these issues, we propose FreDEA, a frequency-aware Dual External Attention framework. FreDEA first uses a globally learned Soft Spectrum Mask to decompose the input into trend and seasonal components, thereby reducing interference among different temporal processes. The trend component is directly modeled by a lightweight linear mapping, while the seasonal component is processed by a Frequency-domain Temporal Learner (FTL), which performs input-dependent soft frequency selection and a complex activation that introduces no additional phase rotation to its immediate input. The Dual External Attention (DEA) module then models temporal and channel dependencies with fixed-size memory projections linear in input length and variable count. Experiments on seven real-world benchmarks show that FreDEA achieves the best performance in 18 out of 35 MSE comparisons and 21 out of 35 MAE comparisons. On the Weather dataset, FreDEA reduces the average MSE by 4.4% compared with the strongest baseline. Ablation studies and visualization results further support the roles of FTL and the two DEA branches under the evaluated settings.

Keywords: Long-term multivariate time series forecasting · Frequency-domain representation · External attention · Series decomposition

1 Introduction

Long-term multivariate time series forecasting aims to predict the future evolution of multiple variables over extended horizons from historical observations, with important applications in power dispatching, traffic control, weather early warning, and macroeconomic planning. Unlike short-term forecasting, long-term observations are usually formed by the complex superposition of macroscopic

trends, multi-period fluctuations, and random noise. Such superposition interferes with the capture of effective long-range dependencies, making it difficult for purely time-domain models to account for both global periodicity and local detailed variations.

To alleviate this problem, decomposition-based forecasting has become a mainstream modeling paradigm. Autoformer [1] embeds seasonal-trend decomposition into deep forecasting, while FEDformer [2] further combines decomposition with frequency enhancement. Frequency-domain methods have also been widely explored because of their ability to characterize global periodicity. FITS [3] formulates forecasting as complex-domain interpolation, FreTS [4] exploits the global perspective and energy concentration of the frequency domain, and FreEformer [5] introduces learnable frequency enhancement into the Transformer framework. Despite this progress, seasonal components are often processed by fixed frequency truncation or globally shared filtering, which cannot adapt to the dynamic spectrum of each input sample. Moreover, phase handling is not always explicit in complex-valued transformations, although phase affects waveform alignment and delay relationships. Therefore, input-adaptive frequency selection and phase-aware processing of seasonal components remain worth further study.

Beyond periodic structures, long-term multivariate forecasting also requires modeling both long-range temporal dependencies and correlations among variables. Transformer models have been widely adopted for relation modeling. PatchTST [6] reduces complexity through patching, while Crossformer [7], iTransformer [8], and SOFTS [9] model multivariate interactions through dependencies across dimensions, variable tokenization, and channel fusion. However, standard self-attention performs pairwise interactions within each sample, leading to quadratic complexity and limited ability to exploit recurring temporal patterns shared across samples, such as daily periodic fluctuations in weather, electricity load, and traffic data. External Attention [10] introduces lightweight shared memory matrices, providing a way to reuse common patterns from the training set. Existing studies such as MEAformer [11] and AFEformer [12] have verified its effectiveness along the temporal dimension, but how to extend shared memory to the channel dimension and integrate temporal and channel dependencies in a reasonable order remains underexplored.

Based on the above analysis, we propose FreDEA, a frequency-aware Dual External Attention framework for long-term multivariate forecasting. Unlike methods that treat frequency-domain processing and dependency modeling as independent modules, FreDEA jointly uses spectral modeling and external attention within a decomposition-based architecture. Specifically, a globally learned soft spectral mask first decomposes the input into trend and seasonal components, where the trend component is directly modeled by a lightweight linear mapping, while the seasonal component is passed to subsequent modules for further refinement. On this basis, the Frequency-domain Temporal Learner performs input-dependent soft frequency selection and applies a modulus activation that preserves the phase of its immediate complex input. Subsequently,

the Dual External Attention module models dependencies in a temporal-first order, where the Temporal External Attention first captures global long-range temporal dependencies, and the Channel External Attention then models inter-variable correlations on top of this temporally enriched representation. The main contributions of this paper are as follows:

- We propose FreDEA, a frequency-aware Dual External Attention framework that unifies adaptive frequency-domain temporal learning and dual external attention within a decomposition-based forecasting architecture.
- We design a Frequency-domain Temporal Learner that refines seasonal representations through input-dependent frequency selection and a complex activation that does not add phase rotation to its immediate input.
- We introduce a sequential TEA-to-CEA dual external attention structure, enabling the model to capture long-range temporal dependencies before modeling inter-variable correlations.
- Extensive experiments on seven real-world benchmarks show that FreDEA achieves the best performance in 18 out of 35 MSE comparisons and 21 out of 35 MAE comparisons. Ablation studies and visualizations support the contributions of FTL, TEA, CEA, and the temporal-first ordering under the evaluated settings.

2 Related Work

2.1 Decomposition and Frequency-Domain Time-Series Modeling

Long-term time series often mix trends and periodic fluctuations, making models in the time domain less effective at capturing global periodicity. Forecasting with decomposition is an important way to alleviate this issue. Autoformer [1] embeds seasonal-trend decomposition into deep forecasting and replaces standard self-attention with an autocorrelation mechanism. FEDformer [2] further combines decomposition with frequency enhancement to improve efficiency and interpretability. Frequency-domain methods provide another natural view of global periodicity: FITS [3] formulates forecasting as complex-domain interpolation, FreTS [4] transfers MLP learning into spectral space, and FreEformer [5] introduces learnable frequency enhancement into Transformer forecasting. Recent models also explore multi-period reshaping, memory projection, graph operations, and dynamic filtering, including TimesNet [13], FiLM [14], FourierGNN [15], FreqLLM [16], JTFNet [17], and FilterTS [18].

These studies demonstrate the value of spectral representations, but two limitations remain. First, many methods still rely on fixed truncation or globally shared filtering, which limits their ability to adapt to sample-specific spectra. Second, phase handling is not always explicit in complex-valued transformations. Complex-valued neural networks [19] suggest that phase is relevant to temporal alignment and waveform reconstruction. In contrast to fixed filtering, FreDEA performs input-dependent frequency selection on the seasonal component and uses a modulus activation that preserves the phase of the activation’s immediate complex input.

2.2 Attention Mechanisms and External Attention

Attention mechanisms are widely used for long-range dependency modeling because they can reorganize temporal and variable information. PatchTST [6] reduces sequence length with patch representations. Crossformer [7] models cross-time and cross-dimension dependencies, iTransformer [8] treats each variable sequence as a token, and SOFTS [9] uses centralized channel fusion for efficient variable interaction. These designs improve multivariate forecasting from different perspectives. Standard self-attention, however, still relies on pairwise interactions within each input sample. Its complexity grows quadratically with sequence length, and it cannot explicitly use recurring patterns shared across samples.

External Attention [10] offers an efficient alternative by replacing dense key-value interactions with small learnable external memory matrices. For a fixed memory size, its memory projections are linear in the dimension being modeled. Recent forecasting models have mainly adopted this idea along the temporal or patch dimension. MEAformer [11] introduces temporal external attention into an MLP forecaster, AFEformer [12] combines temporal external attention with adaptive frequency enhancement, PAM-Net [20] applies patch external attention, and VGANet [21] pairs temporal external attention with graph-based channel modeling. FreDEA differs in applying input-dependent seasonal spectral filtering before sequential temporal and channel external memories. Its contribution lies in this integration and temporal-first ordering rather than in external attention or frequency enhancement in isolation.

3 Methodology

3.1 Problem Definition

This paper studies long-term multivariate time series forecasting. At time step t , the observation is denoted as $x_t \in \mathbb{R}^N$, where N is the number of variables. Given a historical window $X_{t-L+1:t} = [x_{t-L+1}, x_{t-L+2}, \dots, x_t] \in \mathbb{R}^{L \times N}$, the goal is to predict the future sequence $Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \dots, x_{t+H}] \in \mathbb{R}^{H \times N}$. The forecasting objective is to learn a parameterized mapping:

$$\hat{Y}_{t+1:t+H} = f_{\theta}(X_{t-L+1:t}), \quad (1)$$

where $\hat{Y}_{t+1:t+H} \in \mathbb{R}^{H \times N}$ is the prediction and θ denotes learnable parameters. In mini-batch training, the input and target tensors are stacked as $X \in \mathbb{R}^{B \times L \times N}$ and $Y \in \mathbb{R}^{B \times H \times N}$, where B is the batch size. The model is optimized using mean squared error (MSE), and both MSE and mean absolute error (MAE) are used for evaluation. The length dimension L corresponds to temporal dependency modeling, while the variable dimension N corresponds to channel-wise dependency modeling.

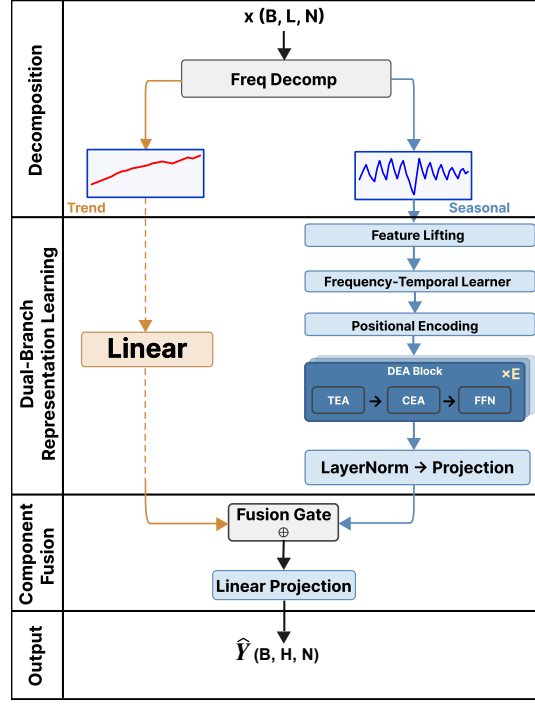


Fig. 1. Overall architecture of FreDEA.

3.2 Overall Architecture

The overall architecture of FreDEA is shown in Fig. 1. FreDEA adopts a dual-branch parallel forecasting architecture: the trend branch captures low-frequency macroscopic evolution with a linear mapping, while the seasonal branch is successively refined by the Frequency-domain Temporal Learner (FTL) and the Dual External Attention (DEA) module for spectral representation enhancement and dependency modeling. At the input stage, Reversible Instance Normalization (RevIN) is applied to mitigate non-stationarity, and the corresponding de-normalization is performed at the output stage to recover the original scale of the forecast.

3.3 Frequency-Aware Component Separation

The first modeling step is to separate low-frequency trends from periodic fluctuations. Given the normalized input in channel-first format $\tilde{X}_c \in \mathbb{R}^{B \times N \times L}$, rFFT is applied along the temporal dimension to obtain $X_f = \mathcal{F}(\tilde{X}_c) \in \mathbb{C}^{B \times N \times F_L}$, where $F_L = \lfloor L/2 \rfloor + 1$. For frequency index $k = 0, \dots, F_L - 1$, the learnable soft spectrum mask is defined as

$$M_k = \text{Sigmoid}(s(c - k)), \quad (2)$$

where c is the soft cut-off frequency and s controls the transition steepness. To keep them in valid ranges, $c = (F_L - 1) \text{Sigmoid}(c_0)$ and $s = \text{Softplus}(s_0) + \epsilon_s$, where c_0 and s_0 are unconstrained learnable parameters. After broadcasting M to the spectrum, the two components are obtained by inverse rFFT:

$$(X_{\text{trend}}, X_{\text{seasonal}}) = (\mathcal{F}^{-1}(X_f \odot M), \mathcal{F}^{-1}(X_f \odot (1 - M))), \quad (3)$$

where both outputs are in $\mathbb{R}^{B \times N \times L}$ and $\mathcal{F}^{-1}$ denotes irFFT. Since c and s are global scalars, this split is shared; input-dependent selection enters later through D_{gate} in FTL.

For the trend component, FreDEA projects the historical length L to the prediction length H by a single linear mapping:

$$Y_{\text{trend}} = X_{\text{trend}} W_{\text{trend}}, \quad (4)$$

where $W_{\text{trend}} \in \mathbb{R}^{L \times H}$, the mapping acts on the last temporal dimension, and $Y_{\text{trend}} \in \mathbb{R}^{B \times N \times H}$. The seasonal component X_{seasonal} is fed into FTL and DEA.

3.4 Frequency-Domain Temporal Learner

After decomposition, the seasonal component still contains useful periodic patterns together with high-frequency disturbances. FTL refines this component in the complex spectral space by selecting informative frequencies, regularizing high-frequency content, and applying a complex activation that adds no phase rotation to the transformed features. As shown in Fig. 2, FTL consists of time-frequency transformation, dynamic spectral masking, spectrum-adaptive dropout, learnable complex-valued transformation, and phase-preserving complex activation.

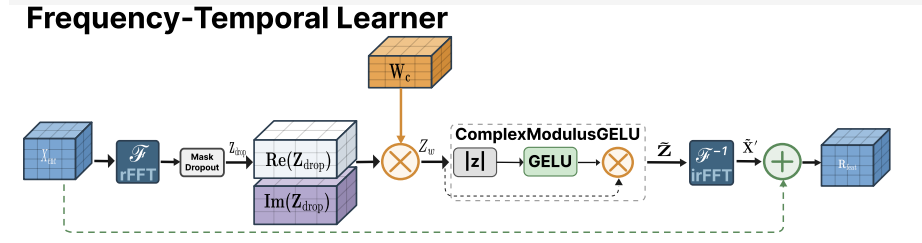


Fig. 2. Structure of the Frequency-domain Temporal Learner.

Feature lifting. Before FTL, X_{seasonal} is lifted to the feature dimension by a learnable vector $e \in \mathbb{R}^d$. Specifically, $X_{\text{enc}}[b, n, \tau, :] = X_{\text{seasonal}}[b, n, \tau] e$, giving $X_{\text{enc}} \in \mathbb{R}^{B \times N \times L \times d}$. The learner then refines X_{enc} in the spectral domain and returns a time-domain representation for residual fusion.

Spectral transform. The input X_{enc} is first rearranged to $X' \in \mathbb{R}^{B \times N \times d \times L}$, where the last dimension corresponds to time. Applying rFFT along this dimension gives $Z = \mathcal{F}(X') \in \mathbb{C}^{B \times N \times d \times F_L}$, with $F_L = \lfloor L/2 \rfloor + 1$.

Adaptive spectral filtering. The dynamic spectral mask combines a learned base prior, a sample-specific gate, and a deterministic high-frequency regularization penalty. The base prior $M_{\text{base}} \in \mathbb{R}^{1 \times 1 \times 1 \times F_L}$ is initialized as a linear decay from 2.0 to δ , where $\delta = -3.0$ if $H \leq L$ and $\delta = 0.0$ otherwise. The dynamic gate is obtained from the magnitude statistic $\bar{m} = \text{Mean}_{N,d}(|Z|)$ by an MLP, yielding $D_{\text{gate}} \in \mathbb{R}^{B \times 1 \times 1 \times F_L}$. For frequency bin k , the regularization penalty is $P_k = 5.0(k/(F_L - 1))^2$. The final spectral mask is

$$M_{\text{filter}} = \text{Sigmoid}(M_{\text{base}} + D_{\text{gate}} - P) \quad (5)$$

with $M_{\text{filter}} \in \mathbb{R}^{B \times 1 \times 1 \times F_L}$. Here, P is a high-frequency prior rather than an aliasing correction, since no resampling occurs. The masked spectrum is $Z_m = Z \odot M_{\text{filter}} \in \mathbb{C}^{B \times N \times d \times F_L}$.

Spectrum-adaptive dropout. To further regularize high-frequency components, FTL applies spectrum-adaptive dropout. The dropout probability for bin k is $p(k) = kp_{\text{max}}/(F_L - 1)$, where $p_{\text{max}} = \min(0.6, p_{\text{base}} \lfloor d/16 \rfloor)$. Let $p = [p(0), \dots, p(F_L - 1)] \in \mathbb{R}^{F_L}$. During training, we sample a frequency mask $K = [K_0, \dots, K_{F_L-1}]$. Each element follows $K_k \sim \text{Bernoulli}(1 - p(k))$, and inverted dropout gives

$$Z_{\text{drop}} = Z_m \odot K \odot \frac{1}{1 - p + \epsilon}, \quad (6)$$

where K and p are broadcast along the B , N , and d dimensions.

Phase-preserving complex activation. The modulus activation is designed not to introduce an additional phase shift after the learnable complex transformation. Let $W_r, W_i \in \mathbb{R}^{1 \times 1 \times d \times F_L}$ and $W_c = W_r + jW_i$. The weighted spectrum is $Z_w = Z_{\text{drop}} \odot W_c \in \mathbb{C}^{B \times N \times d \times F_L}$. Its real and imaginary parts follow standard complex multiplication: $\text{Re}(Z_w) = \text{Re}(Z_{\text{drop}})W_r - \text{Im}(Z_{\text{drop}})W_i$ and $\text{Im}(Z_w) = \text{Re}(Z_{\text{drop}})W_i + \text{Im}(Z_{\text{drop}})W_r$. For any complex element $z \in Z_w$, phase-preserving complex activation is defined as

$$\tilde{z} = z \cdot \frac{\text{GELU}(|z|)}{|z| + \epsilon}. \quad (7)$$

The non-negative real factor preserves the phase of its immediate input z , although W_c may rotate phase. Element-wise activation yields $\tilde{Z} \in \mathbb{C}^{B \times N \times d \times F_L}$.

Temporal reconstruction. The activated spectrum $\tilde{Z}$ is transformed back to the time domain by inverse rFFT and then rearranged to the original layout, producing $\tilde{X}' \in \mathbb{R}^{B \times N \times L \times d}$. Finally, FTL is wrapped by a ReZero residual connection:

$$R_{\text{feat}} = X_{\text{enc}} + \alpha \cdot \tilde{X}', \quad (8)$$

where α is a learnable scalar initialized to 0.

3.5 Dual External Attention

DEA models dependencies after the seasonal representation has been refined in the frequency domain. After positional encoding is added to R_{feat} , the representation is passed through E sequential DEA blocks. Let $U^{(0)} = \text{PosEnc}(R_{\text{feat}})$. The block recursion is

$$U^{(l)} = \text{DEABlock}_l(U^{(l-1)}), \quad l = 1, \dots, E. \quad (9)$$

Each block contains Temporal External Attention, Channel External Attention, and a feed-forward network under pre-normalization. The structures of TEA and CEA are illustrated in Fig. 3 and Fig. 4.

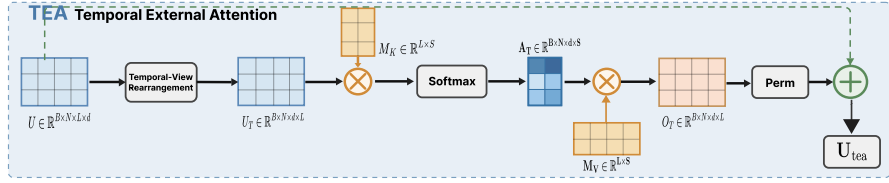


Fig. 3. Temporal External Attention.

Temporal-view External Attention. TEA first models long-range temporal dependencies. Given $U \in \mathbb{R}^{B \times N \times L \times d}$, layer normalization and temporal-view rearrangement produce $U_T \in \mathbb{R}^{B \times N \times d \times L}$. The temporal attention weights are defined as

$$A_T = \text{Softmax}_S(U_T M_K), \quad (10)$$

where $A_T \in \mathbb{R}^{B \times N \times d \times S}$, $M_K \in \mathbb{R}^{L \times S}$ is a shared temporal memory matrix, and Softmax_S normalizes over the external memory dimension. The temporally aggregated output is then

$$O_T = A_T M_V, \quad (11)$$

where $M_V \in \mathbb{R}^{S \times L}$ and $O_T \in \mathbb{R}^{B \times N \times d \times L}$. After permuting O_T back to $\bar{O}_T \in \mathbb{R}^{B \times N \times L \times d}$, the temporal residual update becomes

$$U_{\text{tea}} = U + \text{Dropout}(\bar{O}_T), \quad (12)$$

where $U_{\text{tea}} \in \mathbb{R}^{B \times N \times L \times d}$.

Channel-view External Attention. CEA then models inter-variable correlations on the temporally enriched representation. After normalization, U_{tea} is rearranged to the channel view $U_C \in \mathbb{R}^{B \times L \times d \times N}$. The channel attention weights are defined as

$$A_{\text{ch}} = \text{Softmax}_S(U_C M_K^{\text{ch}}), \quad (13)$$

where $A_{\text{ch}} \in \mathbb{R}^{B \times L \times d \times S}$ and $M_K^{\text{ch}} \in \mathbb{R}^{N \times S}$ is the shared channel memory matrix. The channel-aggregated output is

$$O_{\text{ch}} = A_{\text{ch}} M_V^{\text{ch}}, \quad (14)$$

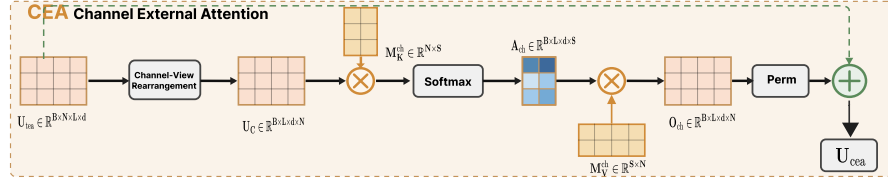


Fig. 4. Channel External Attention.

where $M_V^{\text{ch}} \in \mathbb{R}^{S \times N}$ and $O_{\text{ch}} \in \mathbb{R}^{B \times L \times d \times N}$. Let $\bar{O}_{\text{ch}} \in \mathbb{R}^{B \times N \times L \times d}$ denote O_{ch} permuted back to the original layout. A sigmoid gate $g_{\text{cea}} = \text{Sigmoid}(w_{\text{cea}})$ controls the CEA contribution, with w_{cea} initialized to -5.0 so that the channel branch remains weak before stable temporal representations are formed. The gated residual update is

$$U_{\text{cea}} = U_{\text{tea}} + g_{\text{cea}} \text{Dropout}(\bar{O}_{\text{ch}}), \quad (15)$$

where $U_{\text{cea}} \in \mathbb{R}^{B \times N \times L \times d}$. Finally, let $H_{\text{ffn}} \in \mathbb{R}^{B \times N \times L \times d}$ be the normalized version of U_{cea} . The feed-forward branch forms $V_{\text{ffn}} = \text{Dropout}(\text{GELU}(H_{\text{ffn}} W_1)) \in \mathbb{R}^{B \times N \times L \times d_{\text{ff}}}$ and $O_{\text{ffn}} = V_{\text{ffn}} W_2 \in \mathbb{R}^{B \times N \times L \times d}$, where $W_1 \in \mathbb{R}^{d \times d_{\text{ff}}}$ and $W_2 \in \mathbb{R}^{d_{\text{ff}} \times d}$. The block output is

$$U_{\text{ffn}} = U_{\text{cea}} + \text{Dropout}(O_{\text{ffn}}), \quad (16)$$

where $U_{\text{ffn}} \in \mathbb{R}^{B \times N \times L \times d}$.

3.6 Forecasting Head and Output

After E DEA blocks, the seasonal feature $U^{(E)} \in \mathbb{R}^{B \times N \times L \times d}$ is normalized and permuted to $Q \in \mathbb{R}^{B \times N \times d \times L}$, then projected from length L to horizon H . FreDEA uses linear temporal projection as the base mapping and adds a nonlinear residual term with intermediate length. For any $h \in \mathbb{R}^L$, the projected vector is

$$h_p = h W_p + \gamma_p \cdot \text{Dropout}(\text{GELU}(h W_{p1})) W_{p2}, \quad (17)$$

where $h_p \in \mathbb{R}^H$, $W_{p1} \in \mathbb{R}^{L \times M}$, $W_{p2} \in \mathbb{R}^{M \times H}$, $W_p \in \mathbb{R}^{L \times H}$, and $M = \lfloor (L + H)/2 \rfloor$. The coefficient γ_p is initialized to 0, so the output stage starts from a stable linear temporal mapping and gradually learns nonlinear residuals.

Applying this mapping to the last dimension of Q gives $Q_p \in \mathbb{R}^{B \times N \times d \times H}$, which is permuted to $X_{\text{po}} \in \mathbb{R}^{B \times N \times H \times d}$. The feature branch forms $H_{\text{feat}} = \text{Dropout}(\text{GELU}(X_{\text{po}} W_3)) \in \mathbb{R}^{B \times N \times H \times 2d}$ and $U_{\text{out}} = H_{\text{feat}} W_4 \in \mathbb{R}^{B \times N \times H \times 1}$, where $W_3 \in \mathbb{R}^{d \times 2d}$ and $W_4 \in \mathbb{R}^{2d \times 1}$. Squeezing the last dimension yields $Y_{\text{seasonal}} = \text{Squeeze}(U_{\text{out}}) \in \mathbb{R}^{B \times N \times H}$.

The trend and seasonal predictions are fused by a learnable gate:

$$\hat{Y}_c^{\text{norm}} = g_f Y_{\text{trend}} + (1 - g_f) Y_{\text{seasonal}}, \quad (18)$$

where $g_f = \text{Sigmoid}(w_f)$ and $\hat{Y}_c^{\text{norm}} \in \mathbb{R}^{B \times N \times H}$. Finally, $\hat{Y}_c^{\text{norm}}$ is permuted to $\hat{Y}_{\text{norm}} \in \mathbb{R}^{B \times H \times N}$, and RevIN de-normalization gives the final prediction $\hat{Y} \in \mathbb{R}^{B \times H \times N}$.

3.7 Discussion

The design of FreDEA follows a temporal-first modeling hypothesis: channel interactions may be easier to estimate after temporal context has been enriched. Accordingly, TEA precedes CEA so that channel memory operates on temporally processed representations. The ablation in Sec. 4 shows that reversing this order degrades the average results on all three evaluated datasets, supporting the temporal-first ordering under the evaluated settings without establishing it as universally optimal. The placement of DEA after FTL follows a similar modeling rationale. Spectral refinement is intended to attenuate high-frequency perturbations before dependency modeling. Unlike the globally learned mask used for component separation, the FTL filter adapts to each input sample through D_{gate} . The trend branch remains deliberately simple: its single linear layer provides a low-complexity path for smooth low-frequency movement and limits the capacity assigned to this component. For fixed memory size S , DEA projections cost $O(EBNLdS)$, linear in L and N ; this does not cover FTL, FFN, or the forecasting head.

4 Experiments

This section evaluates FreDEA on seven public long-term forecasting benchmarks, covering main results, ablations, visualizations, and efficiency.

4.1 Datasets

We use four ETT datasets (ETTh1, ETTh2, ETTm1, and ETTm2), Weather, Electricity (ECL), and Exchange. They cover different sampling frequencies, variable numbers, and temporal patterns.

4.2 Baselines

We compare FreDEA with eight representative baselines. Transformer-based baselines include PatchTST [6] and iTransformer [8]; linear baselines include RLinear [22] and DLinear [23]; TimesNet [13] is a convolution-based baseline; S-Mamba [24] and DPME [25] are state-space or Mamba-based baselines; and MSGNet [26] is a graph-based baseline.

Table 1. Multivariate time series forecasting results. **Bold** values denote the best results, and underlined values denote the second-best results among the displayed methods.

Models		FreDEA	DPEM	MSGNet	S-Mamba	iTransformer	RLinear	PatchTST	TimesNet	DLinear										
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE									
ETTh1	96	0.318 0.354	0.326	<u>0.363</u>	<u>0.319</u>	0.366	0.333	0.368	0.334	0.368	0.355	0.376	0.329	0.367	0.338	0.375	0.345	0.372		
	192	0.361 0.378	0.374	0.391	0.376	0.397	0.376	0.390	0.377	0.391	0.391	0.392	<u>0.367</u>	<u>0.385</u>	0.374	0.387	0.380	0.389		
	336	0.390 0.402	0.408	0.415	0.417	0.422	0.408	0.413	0.426	0.420	0.424	0.415	<u>0.399</u>	<u>0.410</u>	0.410	0.411	0.413	0.413		
	720	<u>0.464</u> <u>0.443</u>	0.479	0.454	0.481	0.458	0.475	0.448	0.491	0.459	0.487	0.450	0.454 0.439	0.470	0.450	0.474	0.474	0.453		
	Avg	0.383 0.394	0.396	0.405	0.398	0.411	0.398	0.405	0.407	0.410	0.414	0.407	<u>0.387</u>	<u>0.400</u>	0.400	0.406	0.403	0.405		
ETTh2	96	0.176	0.260	0.173 0.256	0.177	0.262	0.179	0.263	0.180	0.264	0.182	0.265	<u>0.175</u>	<u>0.259</u>	0.187	0.267	0.193	0.292		
	192	<u>0.242</u> <u>0.303</u>	0.247	0.308	0.247	0.307	0.250	0.309	0.250	0.309	0.246	0.304	0.241 0.302	0.249	0.309	0.284	0.362			
	336	0.300 0.340	0.307	0.345	0.312	0.346	0.312	0.349	0.311	0.348	0.307	0.342	<u>0.305</u>	<u>0.343</u>	0.321	0.351	0.369	0.427		
	720	0.394 0.395	0.405	0.402	0.414	0.403	0.411	0.406	0.412	0.407	0.407	<u>0.398</u>	<u>0.405</u>	0.409	0.408	0.403	0.554	0.522		
	Avg	0.278 0.324	0.283	0.328	0.288	0.330	0.288	0.332	0.288	0.332	0.286	0.327	<u>0.281</u>	<u>0.326</u>	0.291	0.333	0.350	0.401		
ETTh1	96	0.371 0.394	0.386	0.405	0.390	0.411	0.386	0.405	0.386	0.405	0.386	<u>0.395</u>	0.414	0.419	<u>0.384</u>	0.402	0.386	0.400		
	192	0.429 0.423	0.443	0.437	0.442	0.442	0.443	0.437	0.441	0.436	0.437	<u>0.424</u>	0.460	0.445	<u>0.436</u>	0.429	0.437	0.432		
	336	0.466 0.441	0.494	0.464	0.480	0.468	0.489	0.468	0.487	0.458	<u>0.479</u>	<u>0.446</u>	0.501	0.466	0.491	0.469	0.481	0.459		
	720	<u>0.494</u> <u>0.461</u>	0.502	0.485	<u>0.494</u>	0.488	0.502	0.489	0.503	0.491	0.481	<u>0.470</u>	0.500	0.488	0.521	0.500	0.519	0.516		
	Avg	0.440 0.429	0.456	0.447	0.452	0.452	0.455	0.450	0.454	0.447	<u>0.446</u>	<u>0.434</u>	0.469	0.454	0.458	0.450	0.456	0.452		
ETTh2	96	0.299	<u>0.345</u>	<u>0.293</u>	0.346	0.328	0.371	0.296	0.348	0.297	0.349	0.288 0.338	0.302	0.348	0.340	0.374	0.333	0.387		
	192	0.377	<u>0.394</u>	0.378	0.399	0.402	0.414	<u>0.376</u>	0.396	0.380	0.400	0.374 0.390	0.388	0.400	0.402	0.414	0.477	0.476		
	336	0.410 0.424	0.421	0.433	0.435	0.443	0.424	0.431	0.428	0.432	<u>0.415</u>	<u>0.426</u>	0.426	0.433	0.452	0.452	0.594	0.541		
	720	0.418	<u>0.437</u>	0.407 0.435	<u>0.417</u>	0.441	0.426	0.444	0.427	0.445	0.420	0.440	0.431	0.446	0.462	0.468	0.831	0.657		
	Avg	0.376	<u>0.400</u>	<u>0.375</u>	0.403	0.396	0.417	0.381	0.405	0.383	0.407	0.374 0.398	0.387	0.407	0.414	0.427	0.559	0.515		
ECL	96	0.148	0.237	0.138 <u>0.236</u>	0.165	0.274	<u>0.139</u>	0.235	0.148	0.240	0.201	0.281	0.181	0.270	0.168	0.272	0.197	0.282		
	192	0.165	0.251	0.158 0.255	0.184	0.292	<u>0.159</u>	0.255	0.162	<u>0.253</u>	0.201	0.283	0.188	0.274	0.184	0.289	0.196	0.285		
	336	0.178	0.267	0.174 0.271	0.195	0.302	<u>0.176</u>	0.272	0.178	<u>0.269</u>	0.215	0.298	0.204	0.293	0.198	0.300	0.209	0.301		
	720	0.225	0.309	0.199 0.295	0.231	0.332	<u>0.204</u>	<u>0.298</u>	0.225	0.317	0.257	0.331	0.246	0.324	0.220	0.320	0.245	0.333		
	Avg	0.179	0.266	0.167 0.264	0.194	0.300	<u>0.170</u>	<u>0.265</u>	0.178	0.270	0.219	0.298	0.205	0.290	0.192	0.295	0.212	0.300		
Exchange	96	0.086 0.208	0.086 0.207	0.102	0.230	0.086 0.207	0.086	<u>0.206</u>	0.206	0.093	0.217	<u>0.088</u>	0.205	0.107	0.234	<u>0.088</u>	0.218			
	192	<u>0.177</u> 0.299	0.179	<u>0.301</u>	0.195	0.317	0.182	0.304	<u>0.177</u>	0.299	0.184	0.307	0.176 0.299	0.226	0.344	0.176	0.315			
	336	0.320	<u>0.413</u>	0.325	<u>0.413</u>	0.359	0.436	0.332	0.418	0.331	0.417	0.351	0.432	0.301 0.397	0.367	0.448	<u>0.313</u>	0.427		
	720	0.869	0.703	0.858	0.698	0.940	0.738	0.867	0.703	<u>0.847</u>	0.691	0.886	0.714	0.901	0.714	0.964	0.746	0.839	0.695	
	Avg	0.363	0.405	0.362	0.405	0.399	0.430	0.367	0.408	<u>0.360</u>	0.403	0.378	0.417	0.367	<u>0.403</u>	0.416	0.443	0.354	0.414	
Weather	96	0.153 0.200	<u>0.161</u>	<u>0.204</u>	0.163	0.212	0.165	0.210	0.174	0.214	0.192	0.232	0.177	0.218	0.172	0.220	0.196	0.255		
	192	0.205 0.247	0.213	<u>0.252</u>	<u>0.212</u>	0.254	0.214	<u>0.252</u>	0.221	0.254	0.240	0.271	0.225	0.259	0.219	0.261	0.237	0.296		
	336	0.260 0.289	<u>0.272</u>	<u>0.295</u>	<u>0.272</u>	0.299	0.274	0.297	0.278	0.296	0.292	0.307	0.278	0.297	0.280	0.306	0.283	0.335		
	720	0.335 0.338	0.352	0.347	0.350	0.348	0.350	<u>0.345</u>	0.358	0.347	0.364	0.353	0.354	0.348	0.365	0.359	<u>0.345</u>	0.381		
	Avg	0.238 0.268	<u>0.249</u>	<u>0.274</u>	<u>0.249</u>	0.278	0.251	0.276	0.258	0.278	0.272	0.291	0.259	0.281	0.259	0.287	0.265	0.317		
1st Count		Total	18	21	8	4	0	0	1	1	1	3	4	3	4	5	0	0	3	0

4.3 Experimental Setup

The historical window length is fixed to $L = 96$, and the prediction horizon is $H \in \{96, 192, 336, 720\}$, following LTSF-Linear and Autoformer. Hyperparameters are tuned for each dataset and horizon. For 96-step to 336-step forecasting, FreDEA uses $d_{\text{model}} = 128$, $d_{\text{ff}} = 256$, dropout from 0.05 to 0.1, batch size 32, and learning rate 0.001. For 720-step forecasting, it uses $d_{\text{ff}} = 12$, dropout 0.3, batch size 128, and learning rate 0.0003. All experiments use PyTorch with Adam on a single NVIDIA RTX 5090 GPU, and early stopping patience is 8.

4.4 Main Results

Table 1 reports results on seven datasets using MSE and MAE. FreDEA ranks first in 18 of 35 MSE comparisons and 21 of 35 MAE comparisons. On Weather, it obtains an average MSE/MAE of 0.238/0.268 and reduces average MSE by 4.4% compared with the strongest baselines, DPEM and MSGNet, both with an average MSE of 0.249. On ETTm2, FreDEA reaches an average MSE of 0.278, lower than DPEM with 0.283 and S-Mamba with 0.288. From horizon 96 to 720, FreDEA’s MSE increases by 0.218, compared with 0.232 for both DPEM and S-Mamba over the same evaluated endpoints.

On Exchange and ECL, FreDEA remains competitive, but its advantage over the strongest baselines is relatively limited. One possible explanation is that weak or non-stationary periodicity in exchange-rate series may reduce the recurrence of temporal patterns across samples, while heterogeneous local dependencies among the many ECL variables may be difficult to represent with a shared channel memory. These interpretations are hypotheses rather than conclusions directly established by the reported experiments.

4.5 Ablation Study

To analyze the main components and the DEA order, we construct five variants: w/o FTL, w/o TEA, w/o CEA, w/o TEA&CEA, and CEA→TEA, which applies channel external attention before temporal external attention. Table 2 reports the results.

Removing FTL raises both MSE and MAE on all three evaluated datasets. For example, ETTm1 MSE increases from 0.383 to 0.390, which is consistent with a contribution from frequency-domain seasonal modeling. Removing TEA or CEA also degrades performance; without CEA, Weather MSE increases from 0.238 to 0.246. Removing both external attention modules further weakens the model. The CEA→TEA variant obtains MSE values of 0.387, 0.443, and 0.243 on ETTm1, ETTh1, and Weather, respectively, all worse than FreDEA. These results support temporal-first ordering for the three datasets and settings included in the ablation.

Table 2. Ablation analysis on ETTm1, ETTh1, and Weather. **Bold** values denote the best result for each metric.

Model	ETTM1		ETTh1		Weather	
	MSE	MAE	MSE	MAE	MSE	MAE
FreDEA	0.383	0.394	0.440	0.429	0.238	0.268
w/o FTL	0.390	0.397	0.443	0.432	0.240	0.270
w/o TEA	0.386	0.395	0.442	0.430	0.240	0.269
w/o CEA	0.387	0.395	0.442	0.431	0.246	0.275
w/o TEA&CEA	0.391	0.398	0.445	0.431	0.246	0.273
CEA→TEA	0.387	0.395	0.443	0.431	0.243	0.272
iTransformer	0.407	0.410	0.454	0.447	0.258	0.278
DPEM	0.396	0.405	0.456	0.447	0.249	0.274
S-Mamba	0.398	0.405	0.455	0.450	0.250	0.276

4.6 Visualization Analysis

We visualize FTL and DEA on one Weather test sample, showing the spectral mask and the external memory responses of TEA and CEA.

Fig. 5(a) shows high spectral responses in low- and middle-frequency regions and weaker responses in high-frequency regions, without fixed-window truncation. Over 10,348 Weather test samples with input length 96 and prediction

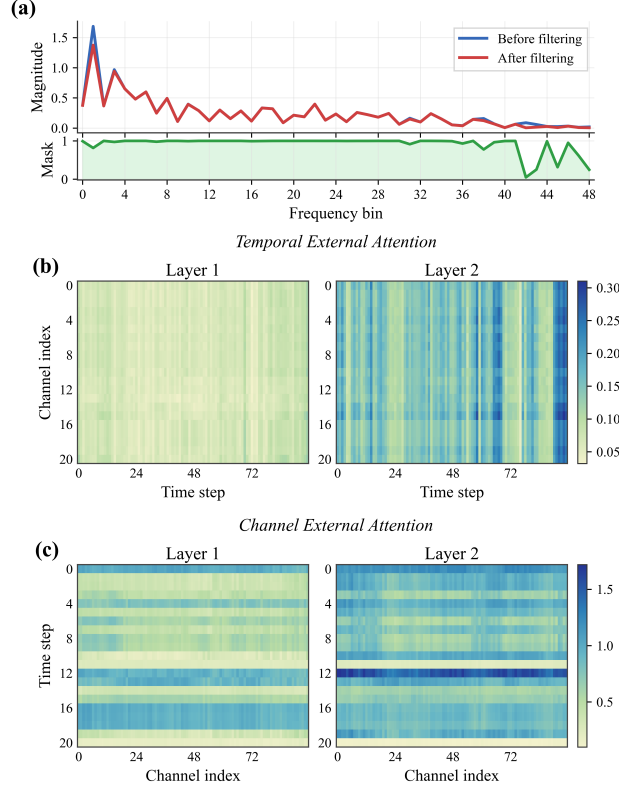


Fig. 5. Visualization of intermediate representations in FreDEA. (a) Spectral mask response. (b) Temporal external attention response. (c) Channel external attention response.

length 192, the average response is 0.9885 for $k < 20$ and 0.5554 for $k > 40$, with a high-frequency standard deviation of 0.3616. This indicates input-dependent soft frequency selection rather than fixed filtering. Fig. 5(b) and Fig. 5(c) show that TEA forms structured temporal responses, while CEA forms concentrated channel responses, matching their roles in temporal and channel dependency modeling.

4.7 Model Efficiency

To further evaluate the computational overhead of FreDEA, this paper selects one representative method from each of five categories of baseline models and measures their parameter counts and single-batch forward inference times under identical input scales.

Table 3. Efficiency comparison under the Weather setting. The input length is 96, the prediction length is 96, the number of variables is 21, and the batch size is 32. Inference time is measured on a single NVIDIA GeForce RTX 5090.

Category	Representative model	Parameters	Inference time
Linear model	DLinear	0.019M	0.140 ms/batch
Transformer model	PatchTST	0.549M	1.319 ms/batch
Convolutional model	TimesNet	1.194M	8.278 ms/batch
Mamba model	DPEM	1.235M	13.718 ms/batch
Graph-based model	MSGNet	3.041M	13.095 ms/batch
Ours	FreDEA	0.231M	4.138 ms/batch

Table 3 compares parameter counts and single-batch inference time using one representative model from each baseline category. FreDEA has 0.231M parameters, fewer than PatchTST, TimesNet, DPEM, and MSGNet. Its 4.138 ms/batch inference time is slower than DLinear and PatchTST because FTL and cascaded TEA-CEA add spectral transforms and two external attention steps, but it remains faster than TimesNet, DPEM, and MSGNet. Thus FreDEA trades modest extra latency over lightweight baselines for higher forecasting accuracy while keeping the parameter scale moderate.

5 Conclusion

This paper presents FreDEA, a frequency-aware Dual External Attention framework for long-term multivariate time series forecasting. Instead of relying on fixed decomposition or intra-sample self-attention alone, FreDEA integrates learned spectral decomposition, input-dependent seasonal filtering, and shared external memory in a unified architecture. For seasonal modeling, the Frequency-domain Temporal Learner uses input-dependent frequency selection and a modulus activation that preserves the phase of its immediate complex input. The sequential TEA-to-CEA structure then captures long-range temporal dependencies before modeling inter-variable correlations. Experiments on seven benchmarks show that FreDEA achieves the best performance in 18 out of 35 MSE comparisons and 21 out of 35 MAE comparisons, with a 4.4% average MSE reduction on Weather over the strongest baseline. Ablation studies, visualization results, and efficiency analysis further support the effectiveness and feasibility of the proposed design. Future work will explore more flexible external memory mechanisms and improve robustness on weakly periodic data and complex heterogeneous variable relationships.

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